AMANOS: An Intent-Driven Management and Orchestration System for Next-Generation Cloud-Native Networks

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The authors introduce an intent-driven system that translates high-level service demands expressed in a natural language and transforms them into continuously monitored deployments.

Abstract

Next-generation mobile networks, namely 5G and beyond, have received significant interest from the research community. This is due to their potential in enabling innovative services and delivering unparalleled performance. To meet these expectations, mobile network operators (MNOs) are investing in cloud native technologies to enhance the reliability, delivery, and adaptability of their services. Consequently, network infrastructures are growing in complexity, particularly with regard to the integration of computing, caching, and communications resources. Managing such complex infrastructure requires novel paradigms that can lower operational costs and enable autonomous management. One promising paradigm for achieving such goals is intent-based networking (IBN). By introducing the notion of intent, IBN empowers systems to autonomously fulfill high-level specified expectations. It emphasizes on the desired outcome rather than the technical details, which proves to be efficient in tackling the escalating network complexity. In this article, we introduce an intent-driven system that translates high-level service demands expressed in a natural language and transforms them into continuously monitored deployments. The proposed system incorporates a closed automation loop that continuously monitors and fulfills the desired expectations. It provides multi-access edge computing (MEC) digital services and 5G core network functions' resources. To validate our approach, we realize a demonstration leveraging a 360° video streaming use case in a physical testbed.

Introduction

In recent years, network infrastructures have been steadily evolving to meet the ever-increasing demands of customers for network services and value-added applications. These new emerging services, such as Virtual Reality (VR), Augmented Reality (AR), and Internet of Everything (IoE), require significant improvements in network latency, reliability, and throughput [1]. To address these demands, new technologies have emerged to support the evolution toward a Beyond 5G ecosystem. One such technology is Network Function Virtualization (NFV), a technology that has revolutionized network architecture. NFV enables the virtual operation of network functions on any general-purpose infrastructure. This has resulted in substantial benefits for MNOs in terms of operational and capital costs. Consequently, MNOs are adopting cloud-native principles and tools for enhanced flexibility and ease of operation [2]. Additionally, MNOs have adopted Multi-Access Edge Computing (MEC) as a computing paradigm to tackle service latency requirements. By placing services at the edge of the network, MEC promotes reduced latency and improved system performance. Nevertheless, the increasing complexity of network infrastructure poses several challenges regarding the management and orchestration of network resources. The dynamic nature of network infrastructures necessitates the involvement of human experts and incurs higher operational costs. Moreover, Service Providers and customers often specify their business requirements at a very high level. Translating these high-level demands into operational tasks on the underlying complex infrastructure poses a significant challenge. In this regard, Intent-Based Networking (IBN) emerges as a promising paradigm that will shape future networks [3].

IBN introduces the notion of "intent" which represents an abstracted operational goal provided as an input to the management system. IBN offers an abstraction layer and emphasizes the desired outcome of the network service, rather than the technical details [3]. This approach proves to be efficient in tackling the escalating network complexity. Future autonomous networks are expected to exhibit a certain level of self-management and self-healing [4]. However, it is imperative to maintain an awareness of the overall objectives and expectations to be achieved. This global guidance is generally referred to as "intent" encompassing expectations in terms of services and constraints, such as energy consumption and budgetary considerations. Intent-based

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systems have proven highly effective in reducing operational costs [5]. These systems are designed to autonomously achieve the goals of the intent, thereby minimizing the need for human intervention. The purpose of intent is to define and communicate knowledge about expectations to a system in a way that allows automated processes to reason about it and derive suitable decisions and actions [4]. An intent can originate from different actors in different formats, depending on the application's context.

While research around IBN is still in its infancy, research contributions are growing extensively and are being validated through proof of concepts [3]. Standardization activities are being held by standards-developing organizations (SDOs), such as the Internet Engineering Task Force (IETF) and TM Forum [3]. Several contributions have been proposed to address various challenges of IBN, such as Intent Translation, Intent Activation, and Intent Assurance [6, 7, 8]. Nevertheless, most of the works don't address the challenge of designing a comprehensive architecture capable of addressing all these processes holistically. In this article, we introduce a complete intent-driven system within the context of next-generation cloud-native networks. The main contributions of this article can be summarized as follows:

- We propose an intent-driven system that leverages Natural Language Understanding (NLU) algorithms to translate customer requirements into an intent model based on ontologies. The intent model is inspired by the recently published TM Forum standards [4].
- The system incorporates a closed automation loop that continuously monitors and fulfills the desired expectations. It enables the provisioning and management of MEC services along with 5G core network functions' resources.
- We design a physical testbed to assess the performance and validate our approach. A 360° video transcoding use case is employed, demonstrating the practicality and efficiency of our system.

The rest of the article is organized as follows: The next section provides background information on intent-based networking and discusses relevant related works. Then we present a general high-level architecture of our proposed intent-driven system. Following that we cover all of the architectural workflows as well as the implementation technologies of the system. Then we outline our experimentation process, including details of the test-bed setup and obtained results. Finally, we conclude by summarizing our work and outlining potential future directions.

Background

This section presents key background knowledge about the IBN paradigm, including relevant definitions, and an overview of related works that address the challenges of realizing intent-driven systems.

Intent-Based Networking

Intent-based networking is a key pillar in enabling next-generation networks. These networks are characterized by their ability to perform self-management, self-healing, and self-operation of resources [5]. However, they still require input from an operator or outside system to provide operational guidance about the goals that the network is to serve [5]. This input and operational guidance are commonly referred to as "*intent.*" A network that allows network operators to provide their input using intent is referred to as an "*Intent-Based Network*" or "*Intent-Driven Network*" [5]. In accordance with the terminology defined in [5], the terms "*intent-driven*" and "*intent-based*" are used interchangeably to describe a system that supports management functions that can be guided by intent. However, generally "intent-based" refers to the interpretation and translation of intent into configurations, primarily associated with the IBN paradigm. On the other hand, "intent-driven" represents a broader perspective that emphasizes that all the system functions are guided through intent. Such a paradigm was the context of several research studies that proposed applications in different fields. This resulted in an important evolution in the definition of intent. The intent is defined by IETF as "*A set of operational goals (that a network should meet) and outcomes (that a network is supposed to deliver) defined in a declarative manner without specifying how to achieve or implement them*" [5]. Recently, TM Forum proposed another definition considered as the latest definition. It describes an intent as "*the formal specification of all expectations including requirements, goals, and constraints given to a technical system.*" Figure 1 illustrates the workflow of an intent-driven network, inspired by the previously mentioned standardization efforts [4, 5]. The role of closed control loops is very important to meet all the requirements specified by customers.

The intent lifecycle starts with customers or end-users specifying their business requirements in a very high-level format to the business layer. The business layer continuously sends reports on the status of the requested services to the clients. Accordingly, the service layer receives an intent that specifies the requested services and the expected quality parameters. This intent is peri-

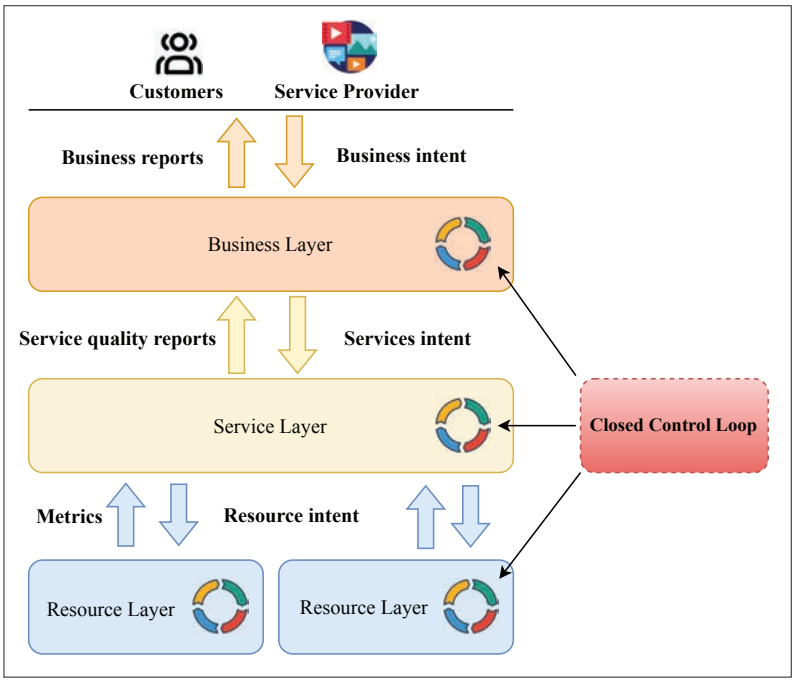


FIGURE 1. High-level workflow of an intent-driven network.

odically monitored and reported to the business layer through quality of service reports. Finally, the service layer sends resource intents to the resource layers specifying the resource allocation strategy. An intent can originate from different operators and vertical entities and can take on diverse forms. It is noteworthy that there are good use cases and benefits for leveraging natural languages [4]. This format offers a degree of flexibility to both customers and operators when specifying their objectives. Simultaneously, it eliminates the need for expertise in a certain format to specify an input to the system. Another advantage of using natural language is the enabling of a chat-based customer experience. However, it is important to acknowledge that this introduces a set of challenges, particularly in the translation of the intent. Natural language is inherently ambiguous and needs interpretation considering context and shared assumptions [4].

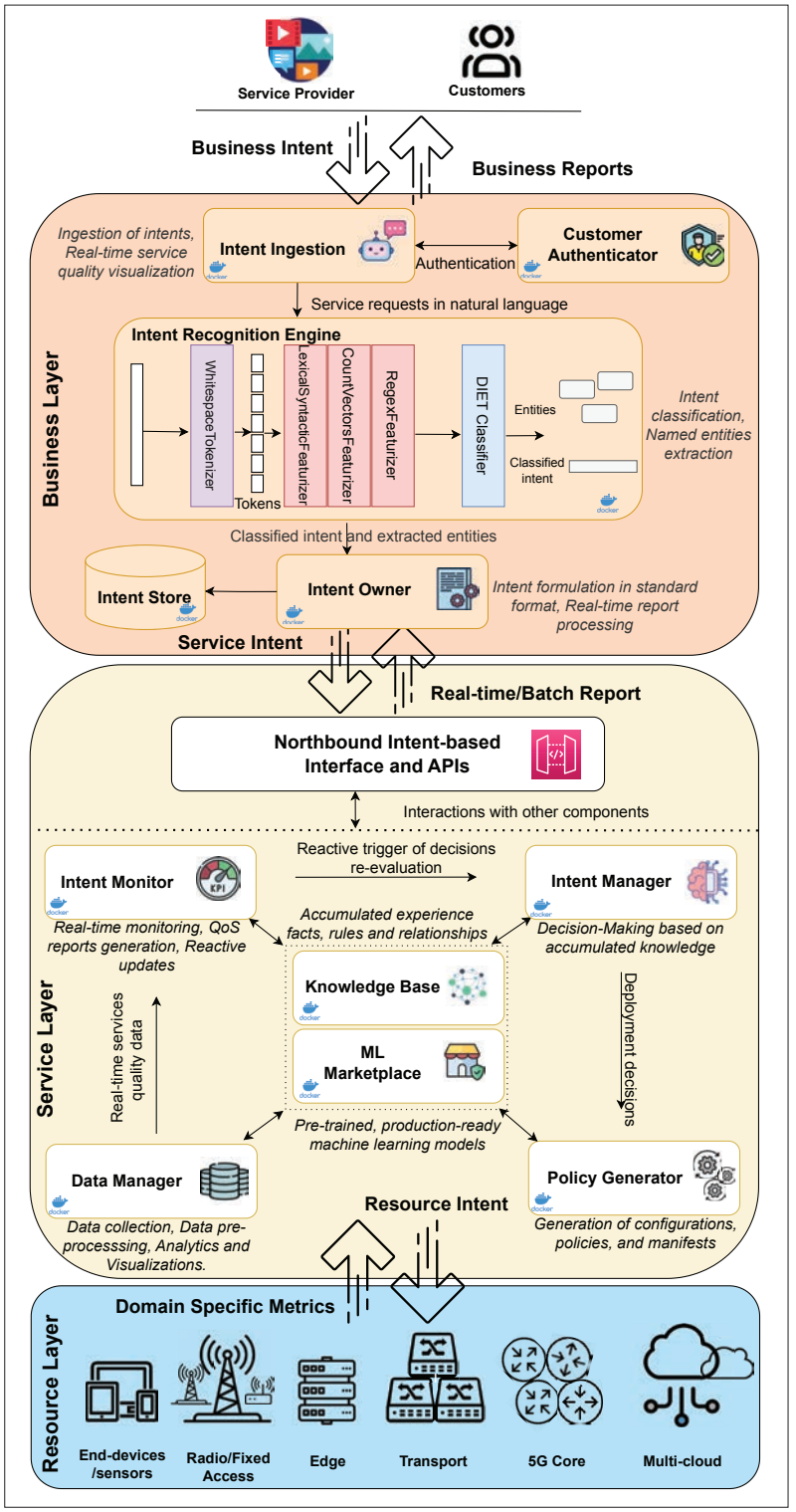


FIGURE 2. AMANOS High-level architecture.

Related Works

Although research and development in the field of IBN are still in their infancy [3]. Few works have been proposed to address the challenges related to achieving intent-driven systems. For instance, authors of [9] have summarized recent research contributions on intent-driven autonomous networks and proposed a general architectural framework. This framework includes an application layer, an intent layer, a network management and orchestration layer, and an infrastructure and resources layer. The authors have reviewed existing solutions for each layer and put forth a set of promising perspectives. In what follows, we provide a comprehensive review of the works realized in the context of IBN, aligning with recently published IBN definitions by SDOs [5]. We focus on the application context, the challenges addressed as well as the added value.

To realize intent-driven systems, a set of components must be designed and integrated. These components include intent profiling, intent translation, intent activation, and intent assurance [5]. Intent profiling serves as the interface through which users express their intents. Intent translation refers to the process of translating customer requirements from an input format to a format understandable by machines [5]. It is a research challenge that has been targeted by authors in [6, 10, 11]. For instance, the authors of [6] proposed a Natural Language Processing (NLP) algorithm based on Multi-Level Perception (MLP). The proposed algorithm translates natural language intent into P4 switch configurations that are enforced on an OpenFlow-based Software Defined Networking (SDN) controller. On the other hand, authors in [10, 11] proposed an NLU algorithm that is based on Named Entity Recognition (NER). The algorithm extracts entities from the user intents and generates a network configuration based on it.

Intent activation is the process of provisioning requested services with the desired parameters [3]. For instance, authors in [12] proposed an intent-based system for service chaining in software-defined Edge-Cloud networks. The received intent is enforced to an SDN controller that executes the generated configurations and creates the chain in an Edge-Cloud environment. In another context, [13] proposed an intent-based fog-native architecture for microservices placement. The proposed architecture includes a workflow manager that receive parsed intents and dynamically compose and deploy microservices workflows. The authors of [14] proposed a high-level architecture that builds on the framework proposed by TmForum. The architecture relies on the collaboration of multiple operators that communicate using different types of intent to achieve the goal of providing services aligned with customer expectations.

Intent Assurance is the process that ensures that the network indeed complies with the desired intent once it has been fulfilled [5]. In this context, [8] proposed an intent-based architecture for video streaming assurance in a MEC-enabled environment. The architecture includes a supervised learning component that estimates the Quality of Experience based on the metrics collected from the video streaming traffic. Furthermore, [7] addressed the challenge of predicting the relationship between the target Key Performance Indicators (KPI) specified in the intent and the current measurements. They proposed an automated deep learning approach that autonomously designs the loss function.

A careful investigation of the aforementioned studies shows that there are few works that consider a complete IBN architecture. Furthermore, there is no standardized IBN architecture for mobile networks. In our work, we target the challenge of designing a complete intent-driven system. We design a set of components joint with their roles and responsibilities. Additionally, we propose algorithms and solutions for the important challenges of IBN. This includes proposing an NLP algorithm that integrates intent classification and NER. We also base our intent activation on a Machine Learning (ML) Marketplace which includes a set of ML models bound to each service to optimize resources. Finally, we leverage cloud-native tools which are a key factor in designing next-generation networks [2]. We evaluate our framework through a physical demonstration comprising a 360° video streaming use case.

AMANOS High-Level Architecture

In this section, we present a high-level architecture illustrated in Fig. 2. The architecture depicts the internal components of the Business, Service, and Resource layers as well as their interactions. In the following subsections, we delve into the responsibility of each component in the architecture and discuss its role in the overall system.

Business Layer

The Business Layer serves as the primary interface between the system and its customers and service providers. It includes an Intent Ingestion component that facilitates the client's interactions with the system [5]. To access the interface, clients are first required to be authenticated with the Customer Authenticator. The customers interact with a ChatBot interface, using natural language to request digital services tailored to their specific requirements. Service providers can also interact with the business layer to request network slices or network functions. The Intent Recognition Engine is responsible for natural language processing. It employs an intent classification and NER model to identify the intent and extract the relevant parameters from the client's request. An important component of the business layer is the Intent Owner [4]. It is responsible for managing the intent by formatting it into a standard structure, saving it, and receiving ongoing reports about service quality and costs. Then the intent owner forwards the intent to the service layer through its exposed interface.

Service Layer

The service layer receives the intents through its northbound interface. These intents, represented using ontologies, express the requested services as well as the expectations. To ensure seamless interoperability, these intents can be formatted in different formats (e.g., RDF, XML, JSON) [4]. The service layer includes a closed control loop represented in Fig. 2, and comprises the Intent Manager (Plan), Policy Generator and Orchestrators (Execute), Data Manager (Monitor), Intent Monitor (Analyze), and the Knowledge Base. This closed-loop draws inspiration from the MAPE-K framework a recognized approach for designing closed loops for autonomous systems. The Intent Manager assumes the responsibility for processing the intent and generating appropriate deployment decisions. These decisions are derived from two primary components: the Knowledge Base, which comprises a collection of facts and rules for fulfilling intents, and the ML Marketplace, which encompasses a set of pre-trained machine learning models ready to use. The intent manager reasons on the received intent as well as the knowledge already acquired to decide how to fulfill the intent. It can decide to leverage the ML marketplace and select an ML model as well as the relevant parameters for deployment. The result of this process is a decision on how to fulfill the intent. Simultaneously, the Policy Generator leverages a set of templates to create configuration files based on the Intent Manager's decisions. These manifests and policies are sent to the orchestrators for activation and deployment. On the other end of the closed control loop, the Data Manager collects real-time metrics data, representing resource usage and KPIs. The Data Manager incorporates automated pre-processing pipelines to clean, filter, and normalize the collected data. An additional primary responsibility of this component is to store all this data (raw and processed) in appropriate formats. The Intent Monitor component is responsible for the continuous monitoring of services. It leverages the metrics and data views sent by the Data Manager and the intent's expectations specified in the Knowledge Base. In case of intent drift, that is, the requested QoS is not satisfied, it triggers a new process that corrects the non-conformity of the deployed service.

Resource Layer

The resource layer represents the infrastructure of an MNO, which can be composed of several domains like RAN, MEC, Core, and Transport. Each domain is managed by a resource orchestrator or controller. Each domain reports metrics about resource usage and service KPIs to be sent to the Data Manager for addressing intent assurance.

Proposed Intent-Driven Management and Orchestration System

Building upon the high-level architecture presented in the previous section, we designed a comprehensive system composed of two primary subsystems. The Customer Service Portal (CSP), and the Autonomous Management and Orchestration System (AMANOS). In what follows, we present the internal workflow of each subsystem, detailing the interactions between each component. Additionally, we introduce the intent model leveraged in our system, which is based on the recent contribution proposed by TM Forum [4].

This format offers a degree of flexibility to both customers and operators when specifying their objectives. Simultaneously, it eliminates the need for expertise in a certain format to specify an input to the system.

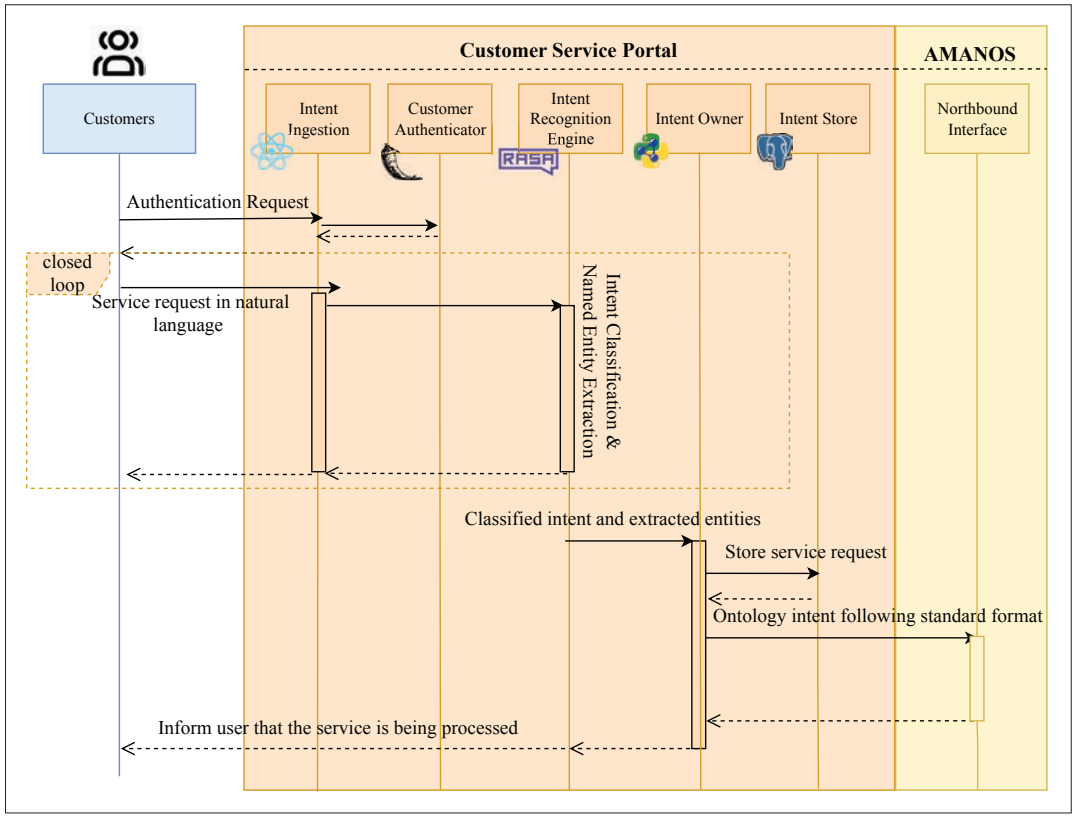


FIGURE 3. Customer service portal's internal components and workflows.

Customer Service Portal

The workflow of the customer service portal and its internal components is illustrated in Fig. 3. The process begins with a client interacting with the *Intent Ingestion* web application which features a user interface for expressing requirements in a natural language. Prior to accessing the web interface, an authentication process is performed by the *Customer Authenticator* component, which handles authorizations for services. The client's requests are processed by the *Intent Recognition Engine,* which is the core component of this subsystem. It is responsible for performing natural language understanding on the received requests provided by the user. These requests are processed using two NLU techniques: Intent Classification (IC) and NER. The IC classifies the client's requests to understand the intent. On the other hand, NER extracts the value entities from customer requests. These entities involve specified service types and quality of service parameters. To implement such a component, we used the Rasa NLU tool and configured the DIET classifier model. The output of the NLU model is then sent to the *Intent Owner*. This component receives the extracted entities to build *Intent-Objects* as *Knowledge Graphs*. The final phase of the workflow concludes by sending the intent to the northbound interface of AMANOS to activate and deploy the intent.

Intent Model

An important research challenge in the context of intent-based networking is intent modeling. Such a challenge has been addressed by the recent TM Forum proposal [4] in which they introduced a common intent model. The proposed intent common model defines high-level classes that can be used to express expectations and relationships. Although the proposed common model includes several classes and properties to express requirements and goals, we found it necessary to extend and customize the model to suit our specific use case. To achieve this, we employed tools such as Rdflib, enabling us to create new classes and properties within the model. The Intent Owner, Intent Manager, and Intent Monitor components leverage this library to interact with the extended intent model. In the following, we present an example of an intent we use in our system, the intent has been translated from the natural language message *"I want to request a 360 video service with a throughput of 10 Mb/s."* Note that this is the received intent by the AMANOS subsystem.

```
        ....
:intent-574116fb a icm:Intent ;
   icm:hasExpectation   :network_
     slice_delivery_expectation,
       :service_delivery_expectation,
       :service_property_expectation.

:service_delivery_expectation a icm:
     DeliveryExpectation ;
   icm:hasParameter :service_type_
     param ;
   icm:hasTarget :serviceTarget .

:service_property0_expectation a
icm:PropertyExpectation ;
   icm:hasParameter :service_through
     put_param ;
   icm:hasTarget :serviceTarget .

:service_throughput_param a icm:
     PropertyParam ;
```

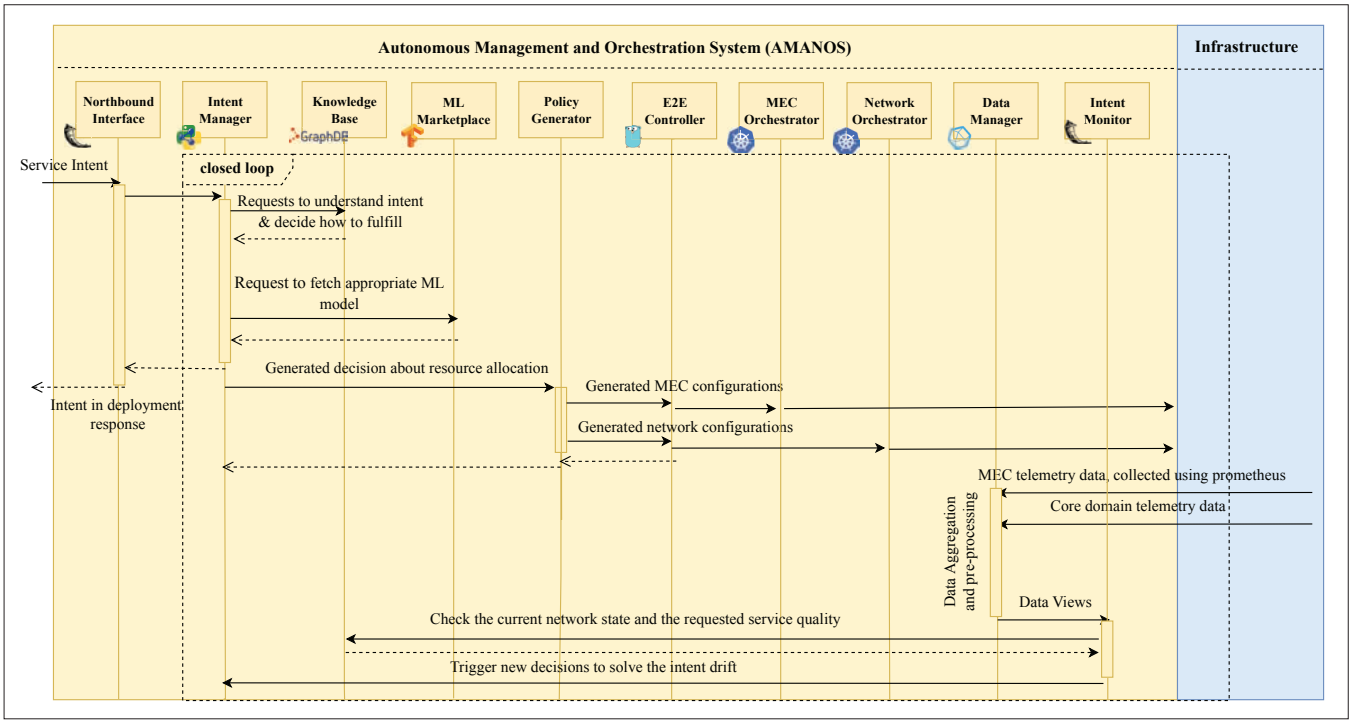


FIGURE 4. Autonomous management and orchestration system's internal components and workflows.

```
    met:throughput[icm:exactly "10"].

:service_type_param a icm:Delivery-
     Param ;
    icm:targetDescription cat:360video.
        ....
```

The intent is formatted in *turtle* format, and it makes use of three namespaces: *icm* represents the intent common model, *met* domain of metrics (e.g., throughput, latency, and reliability for network metrics), *cat* the mobile network operator's services catalog, and *ns* the mobile network operator's network services catalog.

Autonomous Management and Orchestration System

The internal components of the AMANOS subsystem, along with their main workflows and interactions, are illustrated in Fig. 4. The subsystem includes a closed control loop that continuously monitors the underlying services and generates new deployment decisions. The *Northbound Interface* receives standardized intent objects, validates, and then forwards them to the *Intent Manager* for intent activation. The *Intent Manager* processes the received intent as a knowledge graph, employing reasoning to determine the appropriate process to initiate. The *Intent Manager* then fetches the appropriate ML model from the *ML Marketplace* to generate decisions. The ML model selection is based on the service requested and the parameters. The *ML Marketplace* serves production ML models in an h5 Tensorflow format. The generated decisions by the *Intent Manager* are conveyed to the *Policy Generator* to generate configurations. The latter is responsible for mapping these decisions to domain-specific configurations generated from predefined, configurable policy templates. The generated configuration will be sent to the *End-to-End Controller*, which interacts with the underlying domain-specific orchestrators, the *MEC orchestrator*, and the *network orchestrator*. The generated configuration is used by Kubernetes orchestrators to deploy the services. On the other hand, Prometheus is used to collect telemetry data from the underlying infrastructure about service quality and resource usage. The collected data is then streamed to the *Data Manager* component which applies preprocessing on it. In our system, we use InfluxDB which is a Time-Series database that allows efficient processing of data and streaming. The *Data Manager* uses the stored data to create data views for the *Intent Monitor* component. The data views are the result of aggregation and processing of the initial collected raw data. The *Intent Monitor* is therefore responsible for monitoring the intent fulfillment status and generating reactive updates and recommendations. This helps mitigate *intent-drifts* and stay compliant with user and service provider requirements.

Proof of Concept and Evaluation

Experiment Environment

To assess and showcase the effectiveness of our proposed framework, we have established an experimental environment, as illustrated in Fig. 5.

This environment serves as the foundation for a live demonstration of our system, serving as a proof of concept. We leverage a set of Intel NUC (mini computer) to compose a physical Kubernetes cluster. The cluster encompasses both the deployed MEC services as well as the 5G core network functions. The cluster is connected to a personal computer and another Intel NUC where we deploy our system. Finally, we leverage a Raspberry PI 4 to represent user equipment (UE). The personal computer is characterized by an *Intel i7-8750h* processor and *19630 Mib* of memory.

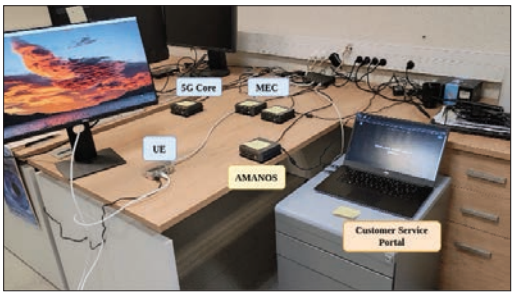


FIGURE 5. Physical testbed environment.

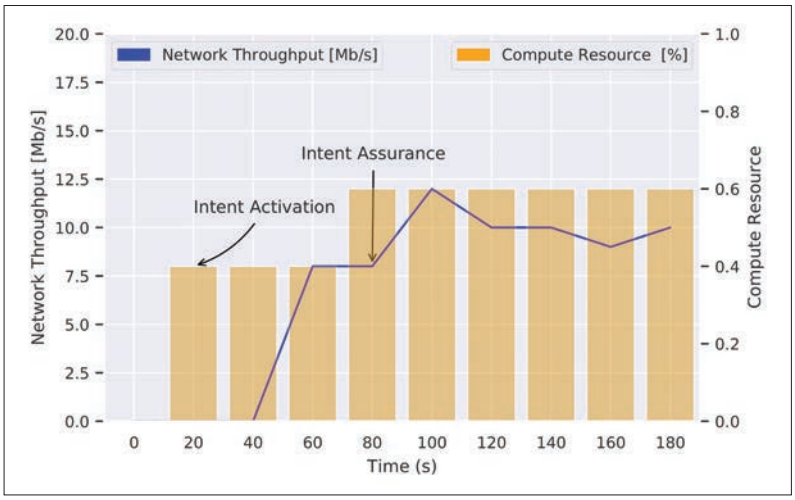


FIGURE 6. 360° video service, network throughput, and resource usage.

On the other hand, the Intel NUC is equipped with an *Intel i5-7300U (2) 3.5 GHz* and a *16 GB* of memory.

Experiment Scenario and Results

The amount of video content delivered over today's mobile network is growing exponentially, and it is predicted that video traffic will account for more than 80 percent of network bandwidth in the future [15]. Meanwhile, AR/VR applications have emerged as one of the most critical use cases in the 5G ecosystem and have been showcased at industry and standards events. Multi-perspective 360° immersive videos have a vital role in implementing AR/VR applications, as they are an initial step in motivating these technologies. To validate our system, it is required that we adopt a use case that is suitable to demonstrate the performance and added value of the system. Hence, we opt for a 360° video application, employing go-transcode, that we package into a Helm chart. Our primary objective with this experiment is to assess the system's efficacy in meeting the expectations outlined in the intent. Given that our focus is on evaluating the entire system, we don't focus on evaluating each component (e.g., NLU algorithm). We believe that evaluating each component on its own is a contribution that has been targeted previously and can be another contribution on its own.

The outcomes of our experiment are visually depicted in Fig. 6, which illustrates the collected network throughput and the percentage of resource utilization throughout the service's lifecycle. The experiment starts at t=0s, with a user (MNO's customer) engaging with our system to articulate their expectations in natural language. The user requests a 360° video service with a 10Mb/s of network throughput. The request is first translated into an intent model represented as ontologies. Subsequently, AMANOS makes resource allocation decisions based on the requested QoS. The resulting configurations are forwarded to the underlying orchestrators for provisioning. At the outset, the service is provisioned using 0.4 percent of computing resources. Once the containers are operational, we start collecting network throughput metrics. These metrics are processed by AMANOS and compared with the expectations specified in the intent. Consequently, a closed-control loop is triggered to rectify the intent-drift situation and generate new allocation decisions aimed at fulfilling the intent. This iterative process ensures that the system automatically strives to align with the user's requested intent and its associated expectations, ultimately resulting in the allocation of 0.6 percent of computing resources as needed.

Conclusion

This article introduces a comprehensive architecture of an intent-driven management and orchestration system tailored to next-generation cloud-native networks. We propose a detailed framework that is aligned with the recent standards in the field. We implement an experimentation testbed to evaluate our system in a 360° video streaming use case. The results of our experiments demonstrate the feasibility of our proposed architecture. Several research directions can be interesting to investigate to enhance the proposed system. For instance, deep reinforcement learning can be a promising approach to integrate within our decision-making system, due to its capacity to manage complex and dynamic environments.

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